

1) Q1: Momentum = mass \* velocity  $\Rightarrow \frac{[M][L]}{[T]}$

Momentum per unit area per unit time which is the momentum flux

Area =  $L^2$

Time =  $T$

$\frac{\frac{[M][L]}{[T]}}{[L]^2 \cdot [T]} = \frac{[M][L]}{[L]^2 [T]^2} = \frac{[M]}{[L][T]^2}$

Force per unit area (which is the stress)

$F = m * a \Rightarrow \frac{[M][L]}{[T]^2}$

$A = L^2$

$\frac{\frac{[M][L]}{[T]^2}}{[L]^2} = \frac{[M]}{[L][T]^2}$

As both have the same dimensions, they are equivalent concepts

Q3: There are two ways momentum can move through a fluid: molecular interactions and the fluid itself.

|                  | Molecular Transport/viscosity                                                                                                                                                           | Convective Transport                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Physical process | Momentum transferred by random particle motion between adjacent fluid layers, all moving at varying velocities. It's a diffusive process that moves downhill along a velocity gradient. | Momentum carried by average motion - known as bulk motion of fluid as it flows from A $\rightarrow$ B. |
| Fluid properties | Viscosity ( $\mu$ )                                                                                                                                                                     | Density ( $\rho$ )                                                                                     |
| Form             | $\tau$                                                                                                                                                                                  | $\rho v v$                                                                                             |
| Comes from       | Velocity gradients like $\frac{dv_x}{dy}$                                                                                                                                               | Bulk velocity like $v_x$                                                                               |

### Problem 1B.1

Viscous stress tensor from Newtonian fluid:  $T_{ij} = -\mu \left( \frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) + \left( \frac{2}{3}\mu - k \right) (\nabla \cdot \mathbf{v}) \delta_{ij}$

Divergence of Velocity = 0

a)  $v_x = by$  ;  $v_y = 0$  ;  $v_z = 0$

$\tau$ : Only non-zero velocity gradient:  $\frac{\partial v_x}{\partial y} = b$

$$\tau_{xy} = \tau_{yx} = -\mu \left( \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right) = -\mu(b+0) = -\mu b$$

$$\tau = \begin{pmatrix} 0 & -\mu b & 0 \\ -\mu b & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\rho v v = \rho (by)^2 = \rho b^2 y^2$

$\rho v v = \begin{pmatrix} \rho b^2 y^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

b)  $v_x = by$  ;  $v_y = bx$  ;  $v_z = 0$

$\tau$ : Non-zero:  $\frac{\partial v_x}{\partial y} = b$  &  $\frac{\partial v_y}{\partial x} = b$

$$\tau_{xy} = \tau_{yx} = -\mu \left( \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right) = -2\mu b$$

$$\tau = \begin{pmatrix} 0 & -2\mu b & 0 \\ -2\mu b & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

c)  $v_x = -by$  ;  $v_y = bx$  ;  $v_z = 0$

$\tau$  : Non-zero  $\Rightarrow \frac{\partial v_x}{\partial y} = -b$  &  $\frac{\partial v_y}{\partial x} = b$

$\tau_{xy} = \tau_{yz} = -\eta \left( \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right) = -\eta(-b+b) = 0$       So  $\tau = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow$  No viscous stresses bc fluid isn't deforming

$\rho v v$  :  $\rho v_x v_x = \rho(-by)^2 = \rho b^2 y^2$   
 $\rho v_y v_y = \rho(bx)^2 = \rho b^2 x^2$   
 $\rho v_x v_y = \rho v_y v_x = \rho(-by)(bx) = -\rho b^2 xy$  }  $\rho v v = \begin{pmatrix} \rho b^2 y^2 & -\rho b^2 xy & 0 \\ -\rho b^2 xy & \rho b^2 x^2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

d)  $v_x = -\frac{1}{2}bx$  ;  $v_y = -\frac{1}{2}by$  ;  $v_z = bz$

-Axisymmetric Stagnation flow radially toward & out of z-axis

$\tau$  : Non-zero  $\Rightarrow \frac{\partial v_x}{\partial x} = -\frac{1}{2}b$   $\frac{\partial v_y}{\partial y} = -\frac{1}{2}b$  &  $\frac{\partial v_z}{\partial z} = b$

$\tau_{xx} = -2\eta \frac{\partial v_x}{\partial x} = -2\eta(-\frac{1}{2}b) = \eta b$   
 $\tau_{yy} = -2\eta \frac{\partial v_y}{\partial y} = -2\eta(-\frac{1}{2}b) = \eta b$   
 $\tau_{zz} = -2\eta \frac{\partial v_z}{\partial z} = -2\eta b$  }  $\tau = \begin{pmatrix} \eta b & 0 & 0 \\ 0 & \eta b & 0 \\ 0 & 0 & -2\eta b \end{pmatrix}$

$\rho v v$  :  $\rho v_x v_x = \rho(-\frac{1}{2}bx)^2 = \frac{1}{4}\rho b^2 x^2$   
 $\rho v_y v_y = \rho(-\frac{1}{2}by)^2 = \frac{1}{4}\rho b^2 y^2$   
 $\rho v_z v_z = \rho(bz)^2 = \rho b^2 z^2$  }  $\rho v v = \begin{pmatrix} \frac{1}{4}\rho b^2 x^2 & \frac{1}{4}\rho b^2 xy & -\frac{1}{2}\rho b^2 xz \\ \frac{1}{4}\rho b^2 xy & \frac{1}{4}\rho b^2 y^2 & -\frac{1}{2}\rho b^2 yz \\ -\frac{1}{2}\rho b^2 xz & -\frac{1}{2}\rho b^2 yz & \rho b^2 z^2 \end{pmatrix}$

2)

Q3: Re for falling film:

$$\frac{4\delta \langle v_z \rangle \rho}{\eta}$$

$\delta$ : film thickness  
 $\langle v_z \rangle$ : avg. velocity

$$\text{Re for flow in circular tube: } \text{Re} = \frac{D \langle v_z \rangle \rho}{\eta}$$

$D$ : Tube diameter

$$\text{Re for flow around sphere: } \text{Re} = \frac{D v_{\infty} \rho}{\eta}$$

$v_{\infty}$ : Approaching velocity of fluid

$$[\text{Re}] = \frac{[D][\langle v_z \rangle][\rho]}{[\eta]} = \frac{[L] \cdot [L/T] \cdot [M/L^3]}{[M/(L \cdot T)]} = \frac{[M L^2 / (L^3)]}{[M / (L T)]} = \frac{[M / (L T)]}{[M / (L T)]} = 1 \quad \text{So Re is dimensionless}$$

Q11: The form isn't possible because at the interface between two immiscible fluids, the tangential component of velocity is continuous—"no-slip" condition—at a liquid's interface. Consequently, fluid A's velocity at the interface must be equal to fluid B's velocity at the interface. According to the sketch, there appears to be a discontinuity in velocity, which doesn't agree with the concept.

2B.2

$$\left. \begin{aligned} \frac{d T_{xz}}{dx} &= \rho g \cos(\beta) \\ T_{xz} &= -\eta \frac{dv_z}{dx} \end{aligned} \right\} \frac{d}{dx} \left( -\eta \frac{dv_z}{dx} \right) = \rho g \cos(\beta)$$

Assuming  $\eta$  is constant:  $\frac{d^2 v_z}{dx^2} = -\frac{\rho g \cos(\beta)}{\eta}$

$$\begin{aligned} \int \frac{d^2 v_z}{dx^2} &= \int -\frac{\rho g \cos(\beta)}{\eta} \\ &= \frac{dv_z}{dx} = -\frac{\rho g \cos(\beta)}{\eta} x + C_1 \end{aligned}$$

At liquid-gas interface  $x=0$  &  $T_{xz}=0$ .

$$0 = -\frac{\rho g \cos(\beta)}{\eta} (0) + C_1 \Rightarrow C_1 = 0$$

$$\begin{aligned} \int \frac{dv_z}{dx} dx &= \int \left( -\frac{\rho g \cos(\beta)}{\eta} x \right) dx \\ v_z(x) &= -\frac{\rho g \cos(\beta)}{2\eta} x^2 + C_2 \end{aligned}$$

At solid wall  $x=\delta$  &  $\overbrace{v=0}^{\text{no slip}}$

$$0 = -\frac{\rho g \cos(\beta)}{2\eta} \delta^2 + C_2 \Rightarrow C_2 = \frac{\rho g \delta^2 \cos(\beta)}{2\eta}$$

$$v_z(x) = -\frac{\rho g \cos(\beta)}{2\eta} x^2 + \frac{\rho g \delta^2 \cos(\beta)}{2\eta} = \boxed{\frac{\rho g \delta^2 \cos(\beta)}{2\eta} \left[ 1 - \left( \frac{x}{\delta} \right)^2 \right]}$$

2B.3

b)  $v_x = v_z(x)$

$$\frac{d\phi}{dz} = \text{constant}$$

Shell:  $\frac{d}{dx}(T_{xz}) = \frac{d\phi}{dz}$

$$\int \frac{d}{dx}(T_{xz}) = \int \frac{d\phi}{dz} = T_{xz}(x) = \frac{d\phi}{dz} x + C_1$$

Since  $x=0$ ,  $C_1=0$

$$\Rightarrow T_{xz}(x) = \frac{d\phi}{dz} x$$

Newtonian fluid

$$T_{xz} = \eta \frac{dv_z}{dx} \quad \text{So} \quad \eta \frac{dv_z}{dx} = \frac{d\phi}{dz} x$$

$$v_z(x) = \frac{1}{2\eta} \frac{d\phi}{dz} x^2 + C_2$$

No slip:  $0 = \frac{1}{2\eta} \frac{d\phi}{dz} B^2 + C_2 \Rightarrow C_2 = -\frac{1}{2\eta} \frac{d\phi}{dz} B^2$

$$v_z(x) = \frac{1}{2\eta} \frac{d\phi}{dz} (x^2 - B^2)$$

$$\text{Max at } x=0 : v_{\max} = v_z(0) = \frac{1}{2\gamma} \frac{d\phi}{dz} B^2$$

$$\bar{v} = \frac{1}{2B} \int_{-B}^B v_z(x) dx = \frac{1}{2B} \cdot \frac{d\phi/dz}{2\gamma} \underbrace{\int_{-B}^B (B^2 - x^2) dx}_{\frac{4}{3} B^3}$$

$$\bar{v} = \frac{1}{2B} \cdot \frac{d\phi/dz}{2\gamma} \cdot \frac{4}{3} B^3 = \frac{2}{3} \cdot \frac{1}{2\gamma} \frac{d\phi}{dz} B^2 = \frac{2}{3} v_{\max}$$

$$\frac{\bar{v}}{v_{\max}} = \frac{2}{3}$$

$$Q = w \int_{-B}^B v_z(x) dx : Q = w \cdot \frac{d\phi/dz}{2\gamma} \cdot \frac{4}{3} B^3 = \frac{2wB^3}{3\gamma} \frac{d\phi}{dz}$$

If  $\Delta P$  over  $L$  drives flow  $\Rightarrow g=0$  so  $\frac{d\phi}{dz} = \frac{-\Delta P}{L}$

$$Q = \frac{-2wB^3}{3\gamma} \frac{\Delta P}{L}$$

2B.7

$$\frac{d}{dr}(r T_{rz}) = 0 \Rightarrow r T_{rz} = C$$

$$T_{rz} = \gamma \frac{dv_z}{dr}$$

$$\gamma r \frac{dv_z}{dr} = C \Rightarrow \frac{dv_z}{dr} = \frac{C}{\gamma r}$$

$$\int \frac{dv_z}{dr} = \int \frac{C}{\gamma r} \Rightarrow v_z(r) = \frac{C}{\gamma} \ln(r) + C_2$$

$$v_z(a) = 0 \quad \& \quad v_z(b) = v_0$$

$$\left\{ \begin{array}{l} \frac{C}{\gamma} \ln(a) + C_2 = 0 \\ \frac{C}{\gamma} \ln(b) + C_2 = v_0 \end{array} \right.$$

$$\frac{C}{\gamma} \ln\left(\frac{b}{a}\right) = v_0$$

$$\text{so } \frac{C}{\gamma} = \frac{v_0}{\ln\left(\frac{b}{a}\right)} = \frac{v_0}{\ln(k)}$$

$$v_z(r) = v_0 \frac{\ln\left(\frac{r}{a}\right)}{\ln(k)} \text{ for } a \leq r \leq b$$

$$b) \quad Q = 2\pi \int_a^b v_z(r) r dr = 2\pi \frac{v_0}{\ln(k)} \int_a^b r \ln\left(\frac{r}{a}\right) dr$$

$$\int_a^b r \ln \frac{r}{a} dr = \frac{b^2}{4} (-2k^2 \ln k + k^2 - 1) \quad ; \quad k = \frac{b}{a}$$

$$Q = \frac{\pi v_0 b^2}{2 \ln(k)} (-2k^2 \ln k + k^2 - 1)$$

$$w = \rho Q \text{ so } Q = \frac{\pi v_0 b^2}{2 \ln(k)} (-2k^2 \ln(k) + k^2 - 1) \text{ for } w = \rho Q$$

c)

$$r=a$$

$$T_{rz} \Big|_{r=a} = \gamma \frac{dv_z}{dr} \Big|_{r=a} = \gamma \cdot \frac{v_0}{\ln(k)} \cdot \frac{1}{a}$$

Total axial viscous force exerted by fluid on rod = shear \* wetted A =  $T_{rz}(a) \cdot 2\pi a L$

$$F = 2\pi a L T_{rz}(a) = 2\pi a L \cdot \gamma \frac{v_0}{\ln(k)} \cdot \frac{1}{a} = 2\pi \gamma \frac{v_0 L}{\ln(k)}$$

$$\ln(k) < 0 \text{ for } k < 1$$

$$F = -\frac{2\pi \gamma v_0 L}{\ln(k)}$$

$$\& \quad |F| = \frac{2\pi \gamma v_0 L}{\ln(k)}$$

↑  
magnitude

d) For narrow annulus, expand expression in exponents of  $\epsilon$ . Leading term simplification:  $F_{\text{slit}} \approx \frac{24V_0 L W}{B}$

$$F = F_{\text{slit}} \left( 1 - \frac{1}{2} \epsilon - \frac{1}{12} \epsilon^2 + \dots + \dots \right)$$

2B.8  $\Delta P = P_1 - P_2 = (\rho_m - \rho_w) g H$

$$Q = \frac{\pi D^4}{128 \eta L} \Delta P$$

Subbing  $\Delta P$ :  $Q = \frac{\pi D^4}{128 \eta L} (\rho_m - \rho_w) g H$

$$1 \text{ kg} = 2.20462 \text{ lb}$$

$$\dot{W}_{\left(\frac{\text{lb}}{\text{hr}}\right)} = \rho_w Q \times 2.20462 \times \overbrace{3600}^{\text{sec}}$$

$$\dot{W} = \rho_w \frac{\pi D^4}{128 \eta L} (\rho_m - \rho_w) g H \times (2.20462)(3600)$$

$$D = 0.10 \text{ in} = 2.54 \times 10^{-4} \text{ m}$$

$$\rho_m = 1594 \frac{\text{kg}}{\text{m}^3}$$

$$\rho_w = 998 \frac{\text{kg}}{\text{m}^3}$$

$$\eta = 1.002 \times 10^{-3}$$

$$g = 9.81 \frac{\text{m}}{\text{s}^2}$$

$$\dot{W} = .004720 \frac{\text{H}}{\text{L}} \frac{\text{lb}}{\text{hr}}$$